2.3 Coasts present opportunities and hazards for people

This chapter will explain:

- ★ the opportunities and hazards of living in coastal areas
- ★ the distribution and characteristics of tropical cyclones and how they can be managed
- ★ detailed specific example: the causes, impacts and potential solutions to coastal erosion in the USA's Eastern Seaboard
- ★ the extent to which coral reefs and mangroves can be managed so that people can benefit from them
- ★ detailed specific example: the Coral Triangle, Southeast Asia.

The opportunities of living near the coast

Coastal areas provide many advantages to people, including health, employment, trade, a place for recreation, tourism and investment, climate. For instance, many people living by the coast experience better health. Living near coastal areas can have a positive impact on mental health and may reduce stress. These areas have long been used for convalescence, holidays and physical activities, such as surfing, swimming, sailing and biking. In addition, the climate in coastal areas is generally milder than in inland areas, with cooler summers and warmer winters.

Coastal areas are also important for employment and trade. Employment opportunities include in the fishing and related industries, such as fish processing, as well as in tourism, with jobs in hotels, restaurants, boating, coastal fishing and water sports. Many coastal areas are important for the import and export of goods, and many ports have developed into major cities, such as London, New York, Shanghai and Singapore.

Coastal areas are also good places for investments (Figure 2.28). In the USA, for example, coastal regions attract residents with higher incomes, and there is a correlation between areas with higher incomes and lower crime rates. This may be because in coastal areas that rely on tourism, residents tend to realise the importance of ensuring the safety and security of visitors, and because residents in coastal areas generally wish to safeguard their investment.

▲ **Figure 2.28** Tourism developments in St Lucia

While coastal areas offer many opportunities to people, their actions may cause problems. These are summarised in Table 2.2.

▼ **Table 2.2** Relationships between human activities and coastal zone problems

Human activity	Agents/consequences	Coastal zone problems
Urbanisation and transport	● Land use changes, e.g. for ports, airports ● Road, rail and air congestion ● Dredging and disposal of harbour sediments ● Water abstraction	● Loss of habitats and species diversity ● Visual intrusion ● Lowering of groundwater table ● Saltwater intrusion ● Water pollution

	● Wastewater and waste disposal	● Human health risks ● Eutrophication ● Introduction of alien species
Agriculture	● Land reclamation ● Fertiliser and pesticide use ● Livestock densities ● Water abstraction	● Loss of habitats and species diversity ● Water pollution ● Eutrophication ● River channelisation
Tourism, recreation and hunting	● Development and land use changes, e.g. golf courses ● Road, rail and air congestion ● Ports and marinas ● Water abstraction ● Wastewater and waste disposal	● Loss of habitats and species diversity disturbance ● Visual intrusion ● Lowering of water table ● Saltwater intrusion in aquifers ● Water pollution ● Eutrophication ● Human health risks
Fisheries and aquaculture	● Port construction ● Fish processing facilities ● Fishing gear ● Fish farm effluents	● Overfishing ● Impacts on non-target species ● Litter and oil on beaches ● Water pollution ● Eutrophication ● Introduction of alien species ● Habitat damage and change in marine communities
Industry (including energy production)	● Land use changes ● Power stations ● Extraction of natural resources ● Process effluents ● Cooling water ● Windmills ● River impoundment ● Tidal barrages	● Loss of habitats and species diversity ● Water pollution ● Eutrophication ● Thermal pollution ● Visual intrusion ● Decreased input of fresh water and sediment to coastal zones ● Coastal erosion

The hazards of living near the coast

As well as opportunities, there are many hazards present in coastal areas. Rising sea levels are threatening to increase flooding in many low-lying coastal zones. Fresh water resources can become contaminated by saltwater intrusion, and some coastal ecosystems, such as salt marshes and wetlands, are being squeezed and eroded. Some coastal areas are at risk from **natural hazards** such as tropical cyclones and tsunamis. **Tropical cyclones** bring high wind speeds (over 119 km/hour) and heavy rain to tropical and sub-tropical coastal areas. **Tsunamis** create large waves that can devastate coastal regions, such as the Indian Ocean tsunami of 2004, in which over 230,000 people died.

Water pollution is another hazard to affect coastal areas. Some of this may be runoff from the land, for example of nitrate fertilisers, that leads to ‘dead zones’ in coastal areas due to the excessive growth of algae using up the oxygen and cutting off light to the lower zones. Some of it may be from the discharge of waste from ships. The discharge of poorly treated wastewater into coastal waters poses environmental and health risks. Moreover, the destruction of natural habitats, such as at Rodney Bay, St Lucia, to create an artificial lagoon (Figure 2.29), may lead to unforeseen impacts. At Rodney Bay currents became stronger, which increased erosion on nearby beaches, and fisheries declined as offshore waters became murkier.

▲ **Figure 2.29** Some tourist developments may lead to unforeseen environmental damage, such as at Rodney Bay, St Lucia

Hard and soft engineering strategies

Human pressures on coastal environments create the need for a variety of **coastal management strategies** (Table 2.3). Coastal defence protects against **coastal erosion** and flooding by the sea. Coastal management strategies may be long term or short term, sustainable or non-sustainable. Successful management strategies require a detailed knowledge of coastal processes. Rising sea levels, more frequent storm activity and continuing coastal development are likely to increase the need for coastal management.

Defence options include:

» '**do nothing**'
» maintaining existing levels of coastal defence
» improving the coastal defence
» allowing retreat of the coast in selected areas.

Hard engineering structures

The effectiveness of sea walls depends on their cost and their performance. Their function is to prevent erosion and flooding, but much depends on:

» whether they are sloping or vertical
» whether they are permeable or impermeable
» whether they are rough or smooth
» what material they are made of (clay, steel or rock, for example).

In general, flatter, permeable, rougher walls perform better than vertical, impermeable, smooth walls.

Cross-shore structures such as groynes, breakwaters, piers and strongpoints have been used for decades. Their main function is to stop the drifting of material. Traditionally, groynes were constructed from timber, brushwood and wattle. However, modern cross-shore structures are often made from rock. They may be part of a more complex form of management that includes beach nourishment and offshore structures.

▼ **Table 2.3** Hard engineering strategies and techniques for coastal management

Type of management	Aim/method	Strengths	Weaknesses
Hard engineering	To control natural processes		
Cliff base management	To stop cliff or beach erosion		
Sea walls 	Large-scale concrete curved walls designed to reflect wave energy	● Easily made ● Good in areas of high density	● Expensive ● Life span about 30–40 years ● Foundations may be undermined
Revetments 	Sloping wooden structures built on shorelines, cliff bases or in front of sea walls to absorb wave energy and reduce coastal erosion	● Easily made ● Cheaper than sea	● Limited life span

		walls	
Gabions 	Rocks held in wire cages to absorb wave energy	● Cheaper than sea walls and revetments	● Small scale
Groynes 	Barriers (usually wooden) placed at right angles to the shoreline designed to trap material transported by longshore drift. They can have a negative impact on coastal zones downdrift as they deprive them of sediment	● Relatively low cost ● Easily repaired	● Cause erosion on downdrift side ● Interrupt sediment flow
Rock armour 	Large rocks at base of cliff to absorb wave energy	● Cheap	● Unattractive ● Small scale ● May be removed in heavy storms
Offshore breakwaters	To reduce wave power offshore	● Cheap to build	● Disrupt local ecology
Rock strongpoints	To reduce longshore drift	● Relatively low cost ● Easily repaired	● Reduces sediment downdrift, which could lead to increased erosion downdrift
Cliff face strategies	To reduce the impacts of sub-aerial processes		
Cliff drainage 	To remove water from rocks in the cliff	● Cost-effective	● Drains may become new lines of weakness ● Dry cliffs may produce rockfalls
Vegetation 	To increase interception and reduce overland runoff	● Relatively cheap	● May increase moisture content of soil and lead to landslides
Cliff regrading 	To lower slope angle to make cliff safer	● Useful on clay (most other measures are not)	● Uses large amounts of land – impractical in heavily populated areas

The sustainability of hard engineering strategies

As the table suggests, all forms of hard engineering can have negative impacts or last only a relatively short time. In that sense, they could be considered to be unsustainable. However, they may protect buildings and businesses and land for many decades, and so in that sense they are extremely important, even though they will need to be replaced or upgraded after a number of years.

Soft engineering strategies

Managed retreat allows nature to take its course – erosion in some areas, deposition in others. Benefits include less money being spent and the creation of natural environments. More soft engineering strategies are outlined in Table 2.4.

The sustainability of soft engineering strategies

Soft-engineering schemes would appear to be more sustainable than hard-engineering schemes as they use, in some cases, natural materials or waste materials. However, soft-engineering also allows land to be eroded and so such schemes may be very unpopular with landowners and homeowners.

The distribution and characteristics of tropical storms and how they can be managed

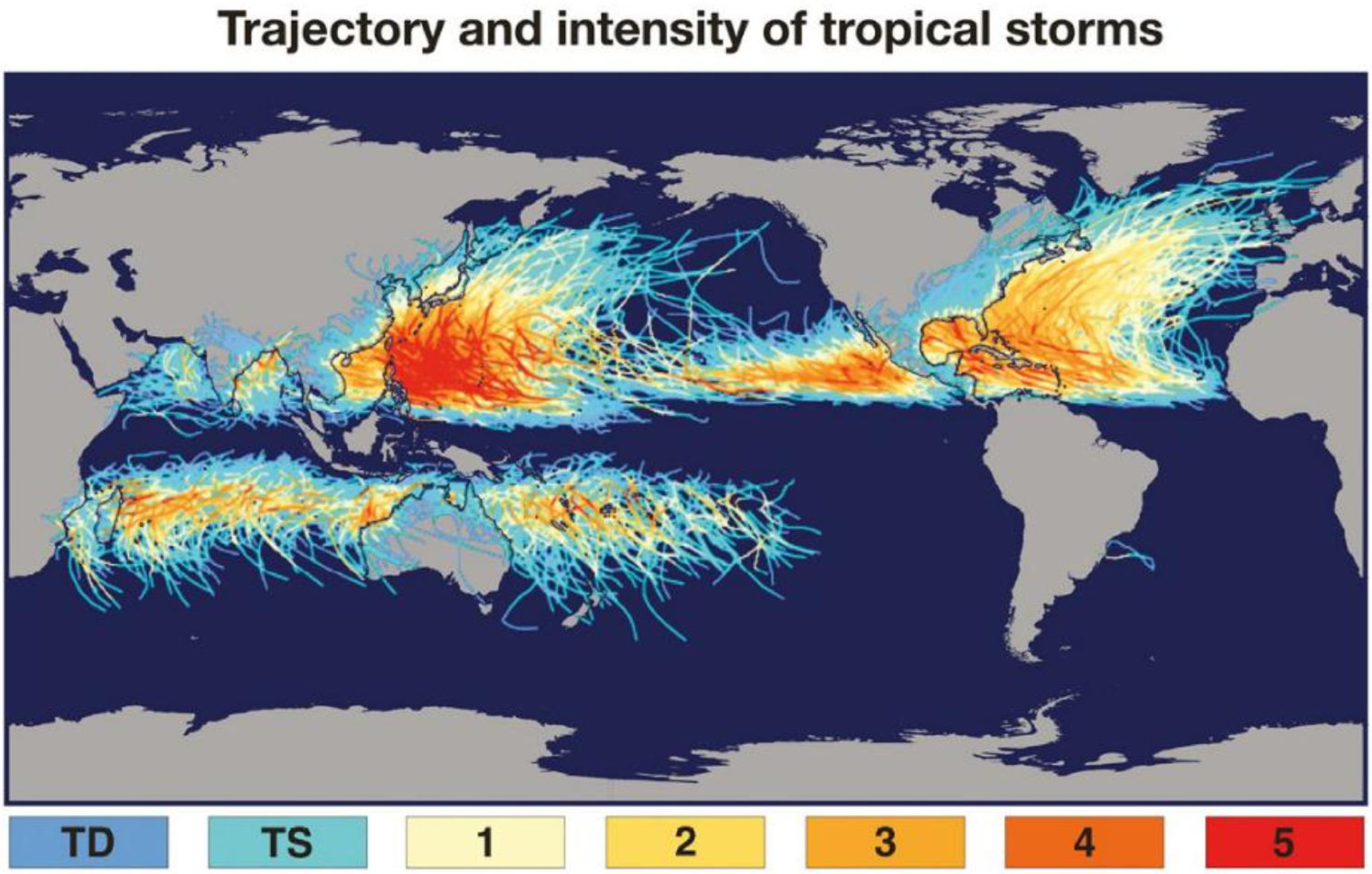
Tropical storms are some of the most dangerous natural hazards for people and for the environment. Damage is caused by high winds, floods and storm surges. In Asia, tropical storms are known as tropical cyclones, and in Northern America they are called hurricanes.

▼ **Table 2.4** Soft engineering strategies and techniques for coastal management

Type of management	Aim/method	Strengths	Weaknesses
Soft engineering	Working with nature		
Offshore reefs	Waste materials, e.g. old tyres weighted down, to reduce speed of incoming waves	● Low technology and relatively cost-effective	● Long-term impacts unknown
Beach nourishment	Sand pumped from sea bed to replace eroded sand	● Looks natural	● Expensive ● Short-term solution
Managed retreat	Coastline allowed to retreat in certain places	● Cost-effective ● Maintains a natural coastline	● Unpopular ● Political implications
'Do nothing'	Accept that nature will win	● Cost-effective	● Unpopular ● Political implications
Red-lining	Planning permission withdrawn New line of defences set back from existing coastline	● Cost-effective	● Unpopular ● Political implications

Interesting note

Tropical storms are known by several different names in different parts of the world. 'Tropical cyclone' is the generic name; 'hurricanes' occur in the Atlantic, 'typhoons' in the western Pacific, and 'cyclones' in the Indian Ocean.

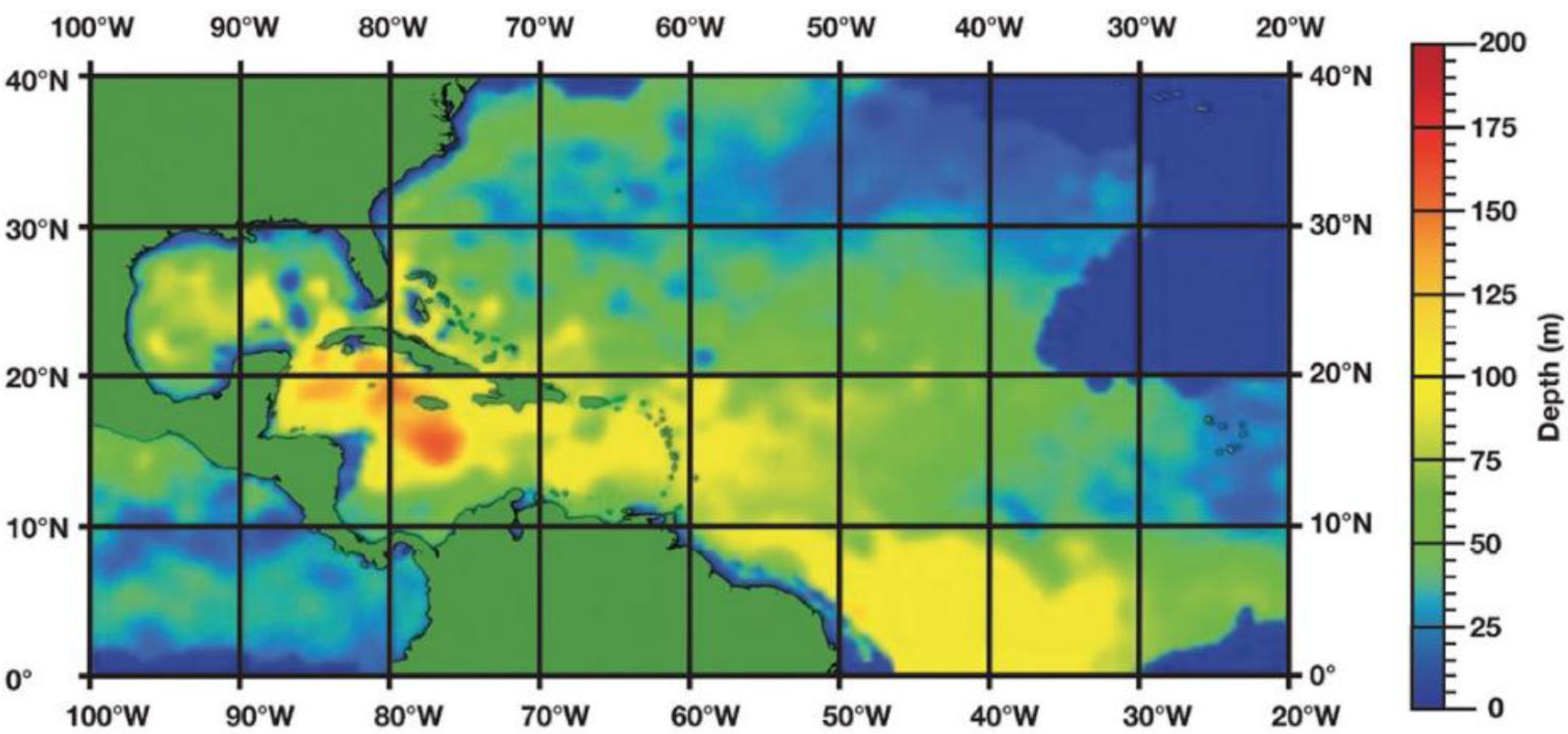


▲ **Figure 2.30** The distribution of tropical cyclones (TD = tropical depression, TS = tropical storm, 1–5 = tropical cyclone categories). The trajectories show the main paths of tropical cyclones 1851–2006.

Tropical cyclones are intense hazards that bring heavy rainfall, strong winds and high waves, and they cause other hazards such as flooding and mudslides. Tropical cyclones are also characterised by enormous quantities of water. This is due to their origin over moist tropical seas. High-intensity rainfall, with totals of up to 500 mm in 24 hours, invariably causes flooding. The path of a tropical cyclone is erratic, so it is not always possible to give more than 12 hours' warning. This is insufficient for proper evacuation measures.

Tropical cyclones develop as intense low-pressure systems over tropical oceans (Figure 2.30). Winds spiral rapidly around a calm central area known as the eye. The diameter of the whole hurricane may be as much as 800 km, although the very strong winds that cause most of the damage are found in a narrower belt up to 300 km wide. In a mature hurricane, pressure may fall to as low as 880 mb. This very low pressure, and the strong contrast in pressure between the eye and the outer part of the hurricane, is what leads to strong gale-force winds.

Tropical cyclones move excess heat from low latitudes to higher latitudes. They normally develop in the westward-flowing air just north of the equator (known as an easterly wave). They begin life as small-scale tropical depressions, localised areas of low pressure that cause warm air to rise. These trigger thunderstorms that persist for at least 24 hours and may develop into tropical storms, which have greater wind speeds of up to 118 km/hr. However, only about 10 per cent of tropical disturbances ever become tropical cyclones – storms with wind speeds above 118 km/hr.



▲ **Figure 2.31** The depth of the Caribbean Sea at which temperatures remain over 27°C

For tropical cyclones to form, a number of conditions are needed:

» Sea temperatures must be over 27°C to a depth of 60 m (Figure 2.31) – warm water gives off large amounts of heat when it is condensed; this is the heat that drives the hurricane.
» The low-pressure area has to be sufficiently far away from the equator so that the Coriolis force (the force caused by the rotation of the Earth) creates rotation in the rising air mass. If it is too close to the equator, there is insufficient rotation and a hurricane will not develop.
» Conditions must be unstable: some tropical low-pressure systems develop into tropical cyclones, but not all of them, and scientists are unsure why some do but others do not.

The impacts of tropical cyclones

The **Saffir-Simpson Scale**, developed by the National Oceanic and Atmospheric Administration, assigns tropical cyclones to one of five categories of potential disaster (Table 2.5). The categories are based on wind intensity, and in order to be classified as a hurricane a tropical cyclone must have maximum sustained winds of at least 118 km/hr. The classification is used for tropical cyclones forming in the Atlantic and northern Pacific – other areas use different scales.

▼ **Table 2.5** The Saffir-Simpson Scale

Tropical cyclone category	Windspeed (km/hr)	Storm surge (metres above normal)	Pressure (mb)	Predicted damage
1	119–53	1.2–1.5	≥ 980	● Minimal damage ● No real damage to building structures ● Damage primarily to unanchored building structures, e.g. mobile homes ● Some coastal road flooding and minor pier damage
2	154–77	1.6–2.5	965–79	● Moderate damage ● Some damage to roofing material, doors and windows ● Considerable damage to vegetation and piers
				● Coastal and low-lying escape routes flood 2–4 hours before arrival of the storm eye ● Small craft in unprotected anchorages break moorings
3	178–209	2.6–3.6	945–64	● Extensive damage ● Some structural damage to small residences and utility buildings ● Flooding near the coast destroys smaller structures, with larger structures damaged by floating debris ● Land below 1.5 m above mean sea level may be flooded 13 km or more inland
4	210–49	3.7–5.5	920–44	

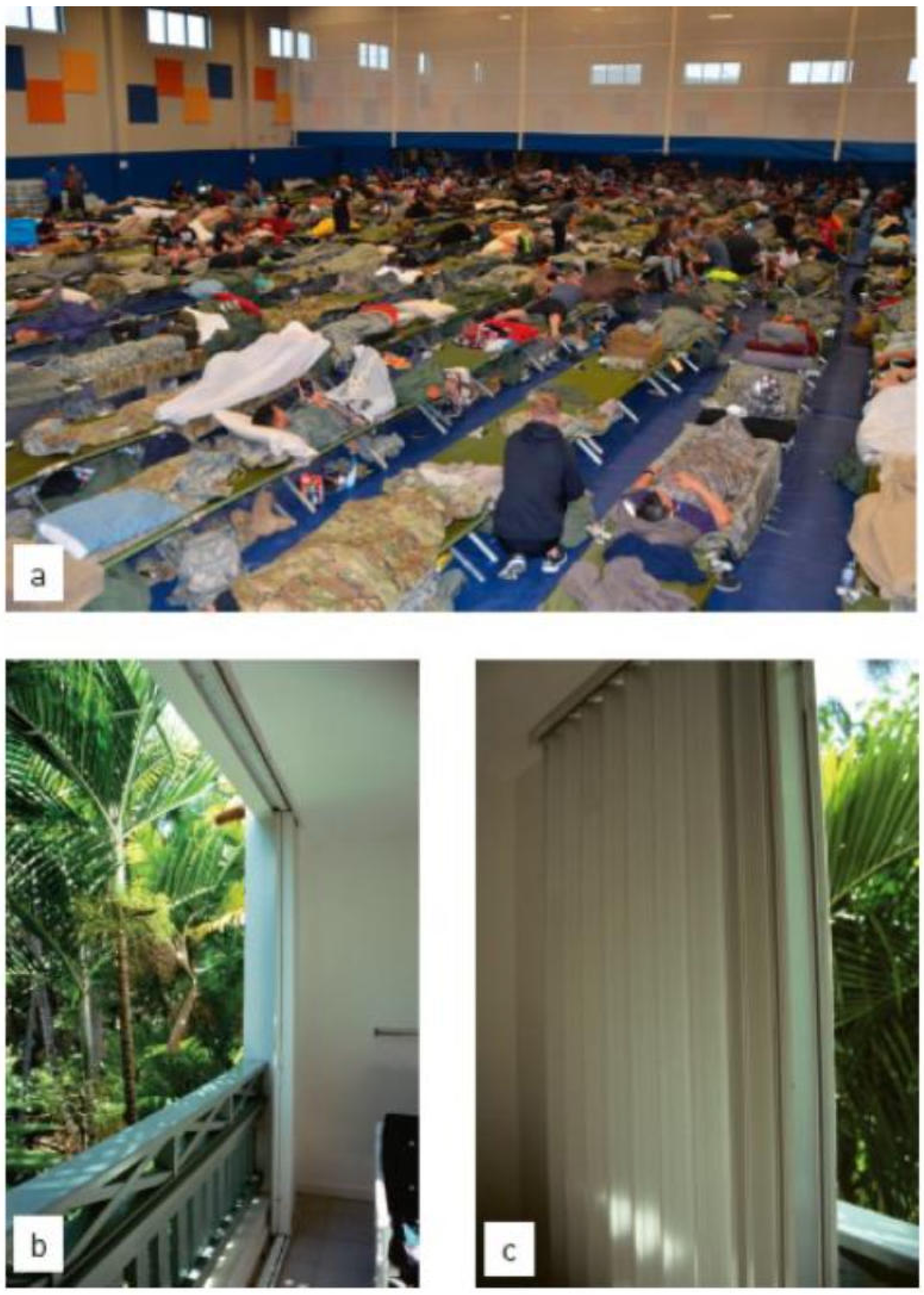
				● Extreme damage ● Some complete roof structure failures on small residences ● Extensive damage to doors and windows ● Land below 3 m above mean sea level may be flooded, requiring massive evacuation of residential areas as far as 10 km inland
5	≥ 250	> 5.5	< 920	● Catastrophic damage ● Complete roof failure on many residences and industrial buildings ● Some complete building failures, with small utility buildings blown over or blown away ● Complete destruction of mobile homes ● Severe and extensive window and door damage ● Low-lying escape routes are cut off by rising water 3–5 hours before the arrival of the centre of the storm ● Major damage to lower floors of all structures located less than 4.5 m above mean sea level and within 500 m of the shoreline ● Massive evacuation may be required of residential areas on low ground within 8–16 km of the shoreline

Other scales exist. The Hurricane Severity Index rates hurricanes (tropical cyclones) on their size and wind speed. For example, a large tropical cyclone that is less intense may cause more damage than a smaller one that is more intense, on account of its size. The National Oceanic and Atmospheric Administration uses the Accumulated Cyclone Energy (ACE), which calculates the estimated maximum sustained velocities at six-hourly intervals. Other scales include the Australian Tropical Cyclone Intensity Scale and the Indian Meteorological Society Scale of Hurricanes.

Predicting tropical cyclones can be difficult for several reasons:

» The unpredictability of hurricane paths makes the effective management of tropical cyclones difficult.
» The strongest storms do not always cause the greatest damage.
» The distribution of the population, for instance throughout the Caribbean Islands, may increase the risk associated with hurricanes.
» Hazard mitigation depends on the effectiveness of the human response to natural events and the protection strategies available (Figure 2.32).
» Developing countries continue to lose more lives to natural hazards as a result of inadequate planning and preparation.

Strategies to manage the impacts of tropical storms



▲ **Figure 2.32** Hurricane management strategies: (a) hurricane shelter; (b and c) steel shutters over windows

Tracking hurricanes

Most hurricanes are tracked by the US National Hurricane Center in Miami, USA, and by the Met Office in the UK. The National Hurricane Center in turn warns countries if they are at risk of a hurricane.

Information regarding hurricanes is received from a number of sources, including:

» satellite images
» aircraft that fly into the eye of the hurricane to record weather information
» weather stations at ground level
» radars that monitor areas of intense rainfall.

Planning and preparing for hurricanes

National governments, international agencies and meteorological societies can help people prepare for a hurricane. They provide advice on risk assessment, land-use control, floodplain management and reducing vulnerability.

» Risk assessment: Risks from tropical cyclones can be shown in a hazard map. The information may be used to estimate the probability of hurricanes hitting a particular place. Weather records may show how often cyclones have struck, and their location and magnitude. In many places records go back for over a century.
» Land-use zoning: This ensures that the most important facilities are placed in the safest areas. Policies regarding future development may need to take into account possible changes related to global climate change.
» Floodplain management: A plan for floodplain management should aim to protect important buildings and infrastructure from river and coastal flooding.
» Reducing vulnerability of structures and infrastructure:
  - New buildings should be designed to be wind and water resistant. Housing is particularly vulnerable to hurricanes. Hurricane Luis (1995) caused damage to 90 per cent of Antigua's houses, while Hurricane Gilbert (1988) made 800,000 people temporarily homeless in Jamaica. To limit damage to houses, owners are now encouraged to fix hurricane straps to roofs and put storm shutters over windows. Buildings can be raised above the ground to protect against floods and storm surges; for instance houses can be built on stilts to allow flood waters to pass away safely below them.
  - Communication and utility lines should be located away from coastal areas or installed underground.
  - Levées and coastal dikes should be regularly inspected for breaches due to erosion.
  - The planting of mangrove trees and other vegetation helps reduce breaking wave energy, and therefore reduces the impact of soil erosion and landslides.

Hurricane watches and warnings

- A tropical storm watch is issued when tropical storm winds are expected within 36 hours.
- A tropical storm warning is issued when there are risks of tropical storm winds within 24 hours.
- A hurricane watch is issued when there is a threat of hurricane conditions within 24–36 hours.
- A hurricane warning is issued when hurricane conditions (winds of 119 km/hr or greater, or dangerously high water and rough seas) are expected within 24 hours.

Keeping safe during a hurricane

Before a hurricane arrives, households should be aware of where the emergency shelters are located, and have some essential supplies at home, such as a first aid kit, essential medical supplies, food and water, portable radio, torch and extra batteries. Windows should be protected with shutters or boarded with plywood. Gardens should be cleared of material that can be picked up and thrown by high winds.

During a hurricane, household members should stay inside, away from windows, skylights and glass doors; listen to the radio or television for hurricane progress reports; store valuables and personal papers in a waterproof container on the highest level of their home; and avoid using open flames as a source of light.

After the hurricane, it is recommended that people seek medical attention for the injured; assist in search and rescue if it is safe to do so; watch out for secondary hazards (fire, flooding, etc.); clean up debris; report damage to utilities; and assist in community response efforts.

The emergency relief offered after a hurricane can take many forms – including providing food supplies, clean water, blankets and medicines. Much of this is provided in hurricane shelters. In some communities, emergency electrical generators may be needed. The community normally becomes involved in the clean-up operation, and electricity and phone companies work to restore power lines and communications.

Long-term redevelopment

Long-term redevelopment may include construction of new buildings in areas away from the coastline and on high ground. Long-term reconstruction concentrates on:

- housing and community projects
- water supply and sanitation
- transport and communications
- agriculture, fisheries and small businesses
- schools
- government expenses.

Activity

To what extent is it possible to manage the impacts of tropical cyclones?

The impacts of Typhoon Haiyan

The Philippines experiences about 20 typhoons every year. In 2012 Typhoon Bopha killed more than 1100 people and caused over US$1 billion in damage.

At least 6000 people were killed in the central Philippine province of Leyte when Typhoon Haiyan, one of the strongest storms ever to make landfall, struck the Philippines in November 2013. The super-typhoon brought winds of up to 315 km/hr and tore roofs off buildings, turned roads into rivers full of debris, and knocked out electricity pylons.

About 70–80 per cent of the buildings in the area in the path of Haiyan in Leyte Province were destroyed. Tacloban, the provincial capital of Leyte, had a population of over 200,000. The storm surge caused sea waters to rise by over 6 m when the typhoon hit. Power was knocked out and there was no mobile phone signal, making communication possible only by radio.

With many provinces left without power or telecommunications, and airports closed in the hardest-hit areas such as Tacloban, it was impossible to know the full extent of the storm’s damage – or to provide badly needed aid. Government figures showed that more than 4 million people had been directly affected. The World Food Programme mobilised some US$2 million in aid and aimed to deliver 40 tonnes of fortified biscuits to victims within days. Estimates of the economic cost are about US$15 billion. Many countries pledged aid to the Philippines, including the UK (US$131 m), Japan (US$52 m), Canada (US$40 m) and the USA (US$37 m).

Satellite images showed normally green patches of vegetation ripped up into brown squares of debris in Tacloban, where a local TV station broadcast images of huge storm surges, flattened buildings and families wading through flooded streets with their possessions held high above the water. Those living in the hardest-hit areas, such as the eastern Visayas, have some of the lowest incomes in the Philippines. Many have little or no savings, so the typhoon put an already vulnerable population at even greater risk of future food and job insecurity. On Bohol Island, where a 7.3 magnitude earthquake had killed some 200 people in the previous month, residents were successfully evacuated ahead of the storm. However, because the island’s main power supply comes from neighbouring Leyte, residents were left without electricity or water. In Tacloban, the sheer force of the storm was just too much for some evacuation centres, which collapsed.

Activity

Describe the main impacts of Typhoon Haiyan.

Detailed specific example

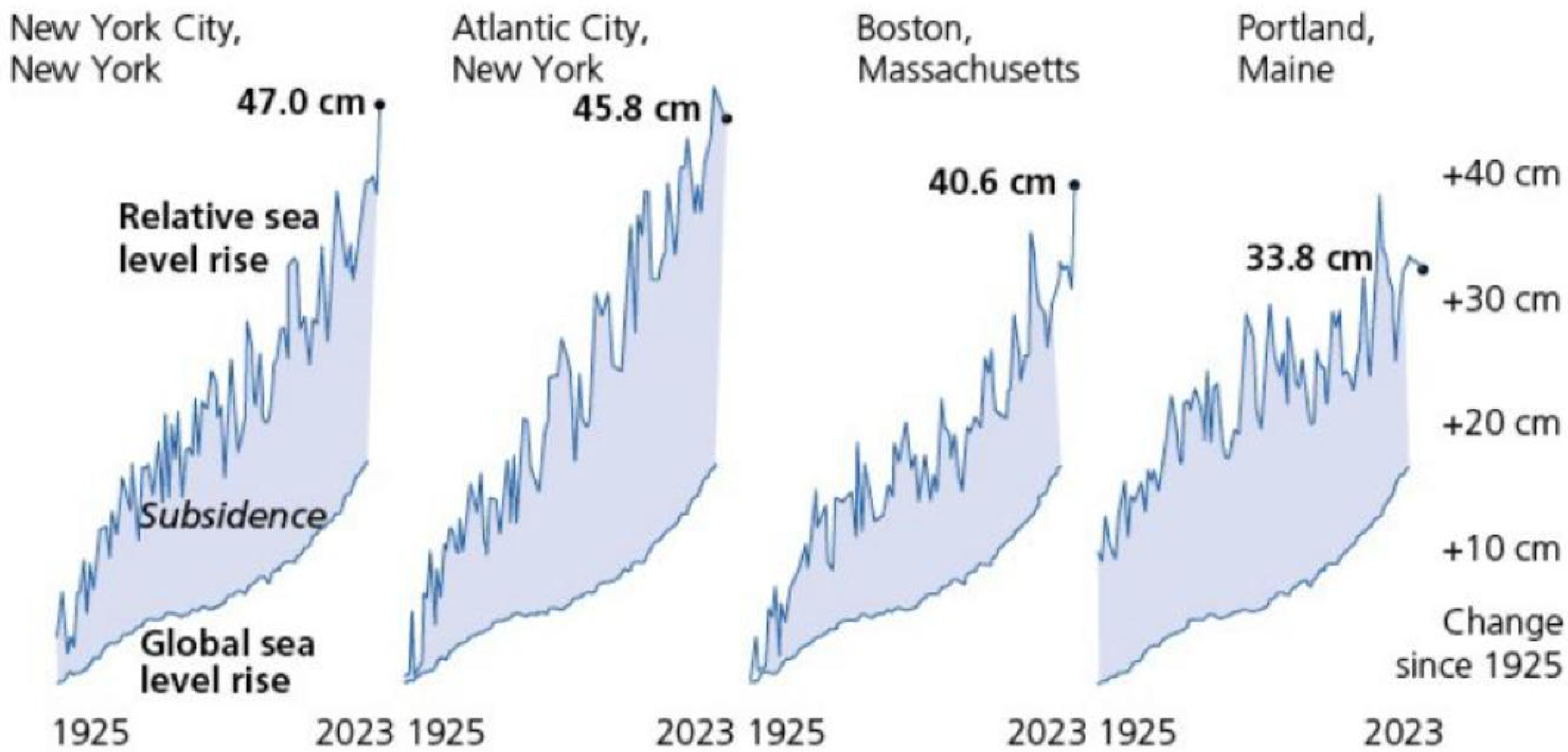
The causes, impacts and potential solutions to coastal erosion in the USA's Eastern Seaboard

Many beaches along the east coast of America have disappeared since 1900, such as that at Marshfield, Massachusetts. Nearly 40 per cent of Americans live along the coasts, where ageing buildings, roads and railways face structural damage from floods. As the sea level rises, the beaches and barrier islands (barrier beaches) that line the coasts of the Atlantic Ocean and the Gulf of Mexico, from New York to the Mexican border, are in retreat.

The problem is that much of the shore cannot retreat naturally because industries and properties worth billions of dollars have been built here. Many important cities and tourist centres, such as Miami, Atlantic City and Galveston (Texas), are sited on barrier islands. Consequently, many shoreline communities have built sea walls and other protective structures to shield them from the power of destructive waves.

- Relief: The flat topography of the coastal plains from New Jersey southward means that a small rise in sea level can allow the ocean to advance a long way inland.
- Changing sea levels: Much of the Northern American coast is sinking relative to the ocean, so local sea levels are rising faster than global averages (Figure 2.33). The level of tides along the coasts shows that subsidence varies between 0.5 and 19.5 mm a year. By contrast, the west coast, in particular in Alaska, is rising. A dominant cause of subsidence along the east coast is groundwater depletion.
- Erosion and tourism-related developments: Extensive coastal development has accelerated erosion as sea level rises, apartment blocks, resorts and second homes have developed rapidly along the shoreline. Erosion is evident at many places along the coasts of the Atlantic and the Gulf of Mexico. Major resorts such as Miami Beach and Atlantic City have pumped in dredged sand to replenish eroded beaches. Erosion threatens islands to the north and south of Cape Canaveral, although the Cape itself appears safe. Resorts built on barrier beaches in Virginia, Maryland and New Jersey have also suffered major erosion.
- Rates of erosion: Overall losses are not well known. Massachusetts loses about 26 hectares (ha) a year to rising seas. Nearly 10 per cent of that loss is from the island of Nantucket, south of Cape Cod. However, these losses are minimal compared with that of Louisiana, which is losing 40 ha of wetlands a day – about 15,500 ha a year.

There have been many attempts to manage the impacts of tropical storms along the USA's southeast coast. However, most homeowners are not covered by insurance for flooding and so have to take measures to protect their properties from flooding. Florida, for example, experiences flash floods from tropical cyclones. Heavy rain typically occurs when a storm makes landfall. The amount of rain can overwhelm drainage systems within minutes. Coastal flooding is also widespread. Low-lying areas and coastal developments are especially vulnerable. Storm surges can also add to flooding. Cities in Florida that are vulnerable to flooding include Miami, Pensacola, Tampa and Fort Lauderdale.



▲ **Figure 2.33** Sinking land and rising sea along the east coast of the USA

Protective measures include the use of trap bags (a type of modified sand bag) to set up flood barriers, and levées. Sea walls have been built in many locations. However, these may give protection only up to a certain magnitude of event. With global climate change, storm heights may become greater and existing sea walls may prove ineffective. Temporary dams can be built using traditional sand bags. Sand bags and trap bags can be used to prevent beach erosion.

Following the devastation of Hurricane Andrew, a Category 5 Hurricane in 1992 that killed 65 people and caused over $27 billion worth of damage, Florida's Division of Emergency Management created the Hurricane Loss Mitigation Program to provide retrofits to residential, commercial and mobile homes. The Gulf State College Mobile Home Tie-Down Program inspects and provides tie-downs for mobile homes to protect them against the high wind speeds experienced in tropical storms.

According to the Canadian engineering company WSP, current rates of subsidence are increasing the risk of sea level change. Up to 2.1 million people are affected by subsidence annually in 867,000 properties. Hot spots include New York, Florida and Charleston, South Carolina. WSP invested in evacuation routes and access to hospitals. They have raised roads at Key Largo, Florida and Virginia Beach, Virginia. The North Carolina Department of Transport has managed the risk of flooding by raising parts on Interstate 40 and Highway 53.

Activities

1 Explain why beaches on the Eastern Seaboard of the USA are retreating.
2 State the proportion of Americans who live along the coast.
3 Estimate by how much the east coast is subsiding each year.

The extent to which coral reefs and mangroves can be managed

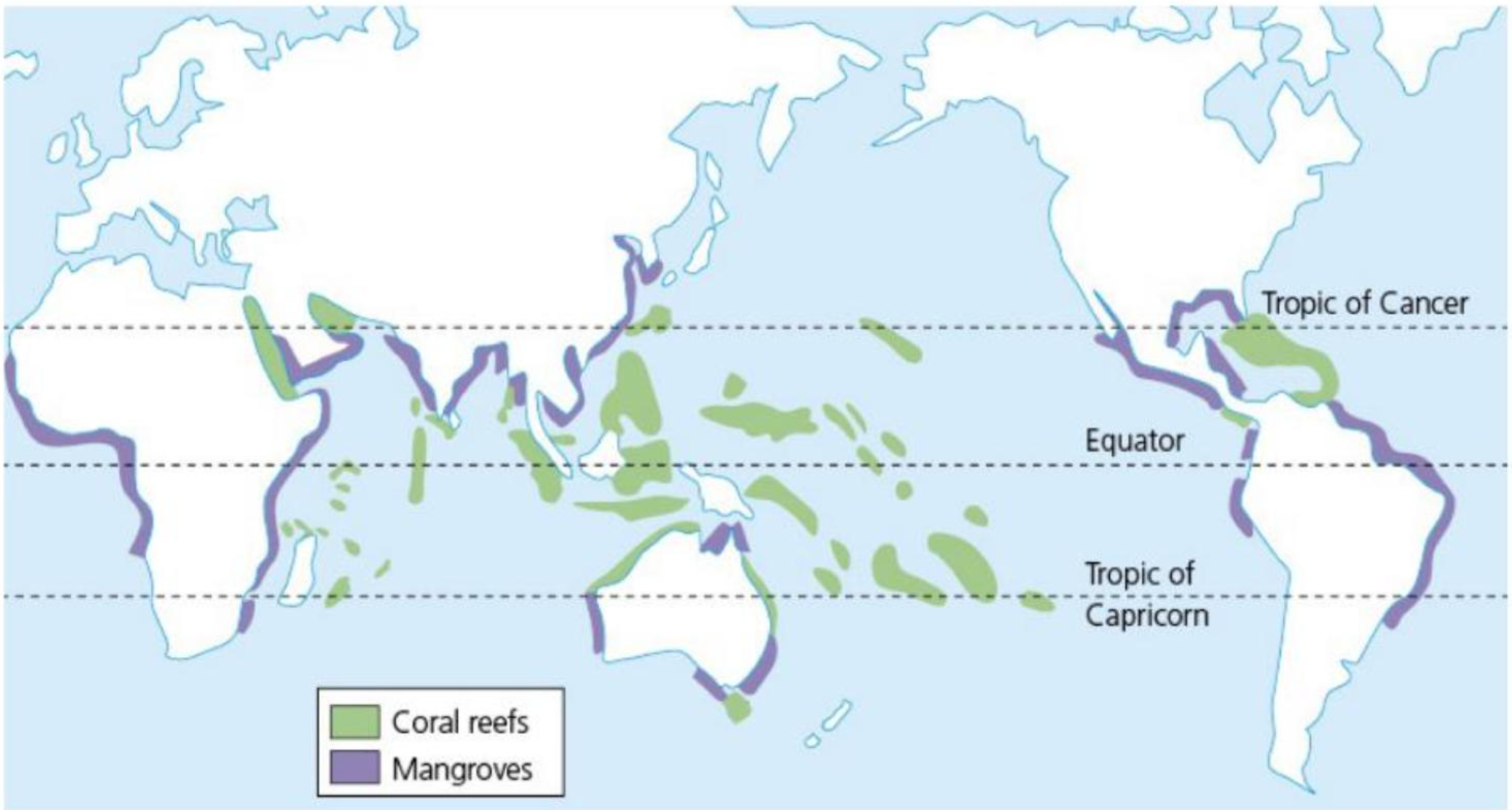
Coral reefs and **mangroves** are other types of coastal ecosystems that people can benefit from (Figure 2.34).

Coral reefs

Coral reefs are generally located in tropical seas. They are calcium carbonate structures, made up of reef-building stony corals. Coral is limited to the depth that light can reach, so reefs develop in shallow water, ranging to depths of 60 m. This dependence on light also means that reefs are only found where the surrounding waters contain relatively small amounts of suspended material. Reef-building corals live only in tropical seas, where temperature, salinity and clear water allow them to develop.

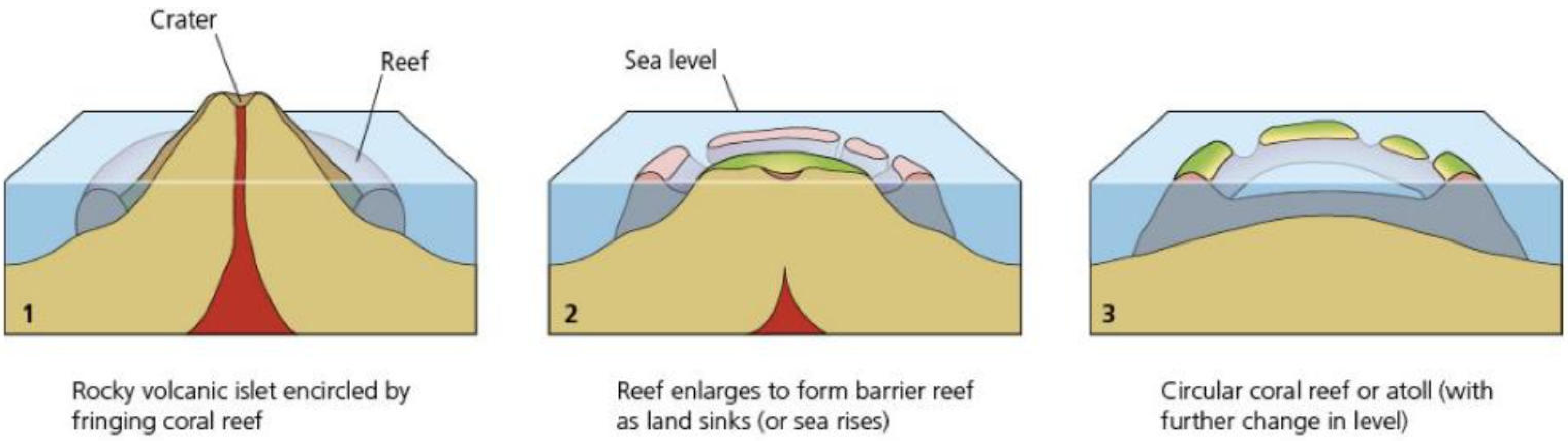
There are many types of coral reef (Figure 2.35):

» Fringing reefs are those that fringe the coast of a landmass (Figures 2.36 and 2.37). Many fringing reefs grow along shores that are protected by barrier reefs and are thus characterised by organisms that are best adapted to low wave-energy conditions.



▲ **Figure 2.34** The distribution of coral reefs and mangroves

» Barrier reefs occur at a greater distance from the shore than fringing reefs and are commonly separated from it by a wide, deep lagoon. Barrier reefs tend to be broader, older and more continuous than fringing reefs. For example, the Beqa barrier reef off Fiji stretches unbroken for more than 37 km, and that off Mayotte in the Indian Ocean for around 18 km. The largest barrier reef system in the world is the Great Barrier Reef, which extends 1600 km along the east Australian coast, usually tens of kilometres offshore. Another long barrier reef is located in the Caribbean, off the coast of Belize between Mexico and Guatemala.
» Atoll reefs rise from submerged volcanic foundations. Atoll reefs are essentially indistinguishable in form and species composition from barrier reefs, except that they are confined to the flanks of submerged oceanic islands, whereas barrier reefs may also flank continents. Over 300 atolls are present in the Indo-Pacific, but only ten are found in the western Atlantic.



▲ **Figure 2.35** The formation of coral reefs

▲ **Figure 2.36** A fringing reef on the south coast of Antigua

▲ **Figure 2.37** A fringing reef on the west coast of Antigua

Coral reefs are often described as the 'rainforests of the sea' on account of their rich biodiversity. Some coral is believed to be 2 million years old, although most is less than 10,000 years old. Coral reefs contain nearly a million species of plants and animals, and about 25 per cent of the world's sea fish breed, grow and evade predators in coral reefs. Some of the world's best coral reefs include Australia's Great Barrier Reef, many of the reefs around the Philippines and Indonesia, Tanzania and the Comoros, and the Lesser Antilles in the Caribbean.

Coral reefs are of major biological and economic importance. Countries such as Barbados, the Seychelles and the Maldives rely on tourism based on their reefs. Florida's reefs attract tourism worth over $1 billion annually and create more than 70,000 jobs. The global value of coral reefs in terms of fisheries, tourism and coastal protection is estimated to be US$650 billion.

Threats to coral reefs

Coral reefs face many pressures, for example from the fishing industry. The industry uses dynamite to flush out fish and cyanide solution to catch live fish. Destruction of the reefs also takes many other forms – from the collection of specimens, trampling, berthing of boats, oil spills, mining and the cement industry. Indirect pressures include sedimentation from rivers and waste disposal from urban areas. Coastal development, especially for tourism, is taking its toll too. Dust storms from the Sahara have introduced bacteria into Caribbean coral, while global warming may cause coral bleaching. Bleaching occurs when high temperatures kill the algae in coral, removing their colour so the coral appears bleached. Many areas of coral in the Indian Ocean were destroyed by the 2004 tsunami.

Floating plastic pollution blocks out some sunlight, spreads diseases and cuts into coral, making it more susceptible to disease. Studies of more than 150 coral reefs in Australia, Indonesia, Malaysia and the Philippines showed that when coral came into contact with plastic, the likelihood of disease increased from 4 per cent to almost 90 per cent. In addition, coral is very vulnerable to oil pollution and runoff from fertilisers. Runoff from fertilisers or sewage can be particularly damaging, as coral reefs are adapted to low nutrient levels.

Overfishing is one of the biggest threats to coral reefs; the other is global climate change. It is believed that if atmospheric carbon dioxide levels rise to 450 ppm (they were over 420 ppm in 2024), coral reefs could die off. This would affect the lives of the up to 500 million people who depend on coral, and reduce the US$100 billion that coral provides to the human economy. Global climate change and other human activities have affected about 20 per cent of the world's coral. Global temperatures may rise by 2°C, which could lead to widespread coral bleaching, extinction of coral species, more fragile coral skeletons and greater risk of storm damage.

Sustainable techniques to protect coral reefs

To avoid permanent damage and to support people in the tropics, it is recommended that:

- the world community reduces emissions of greenhouse gases and develops plans to **sequester** carbon dioxide
- damaging human activities, including overfishing, blasting coral and sedimentation, are halted to allow coral reefs to recover
- more coral reefs are designated as Marine Protected Areas (MPAs) to act as reservoirs of biodiversity, including many remote and uninhabited reefs that are still in good condition
- the management, monitoring and enforcement of regulations is improved
- local coastal management practices are introduced
- assistance is provided to LICs and MICs
- alternative lifestyles are developed that reduce pressure on coral reefs.

Activities

1. Identify the conditions in which coral grows.
2. State the difference between a fringing reef and a barrier reef.
3. Explain how atolls are formed.
4. Explain why coral reefs are so valuable.
5. Examine the main threats to coral reefs.

Mangroves

Mangroves are salt-tolerant forests of trees and shrubs that grow in the tidal estuaries and coastal zones of tropical areas (Figure 2.38). The muddy waters, rich in nutrients from decaying leaves and wood, are home to a great variety of sponges, worms, crustaceans, molluscs and algae. Mangroves cover about 25 per cent of the world's tropical coastline, the largest being the 570,000 ha Sundarbans in Bangladesh.

▲ **Figure 2.38** Mangrove swamps

Mangroves have many uses, such as providing large quantities of food, fuel, building materials and medicine. One hectare of mangrove in the Philippines can yield 400 kg of fish and 75 kg of shrimp. In addition, mangroves protect coastlines by absorbing the force of tropical cyclones and storms. They also act as natural filters, absorbing nutrients from farming and sewage disposal.

Mangroves and coral reefs are fundamentally connected ecosystems. Mangroves protect coral reefs from sedimentation from land-based sources, as well as helping to keep the water clear of particles and nutrients. Both of these functions are necessary to maintain reef health. Mangroves also provide spawning and nursery areas for many animal species that spend their adult lives on the reefs. In return, the coral reefs provide shelter for the mangroves and their inhabitants, while the calcium carbonate eroded from the reef provides sediment in which the mangroves grow.

Despite their value, many mangrove areas have been lost to rice paddies and shrimp farms. Between 2000 and 2016, 20–35 per cent of global mangrove forests were lost, mainly due to human activities but some due to erosion. Over 60 per cent of mangrove losses were due to conversion to aquaculture and agriculture. One in six species found in mangrove ecosystems is now threatened with extinction. Table 2.6 shows regional variations in the cause of mangrove losses, 2000–16. As population growth in coastal areas is set to increase, the fate of mangroves looks bleak.

Other threats to mangroves include:

- clearance for coastal developments, for example hotels
- deforestation for wood and charcoal
- browsing (feeding on higher plants) by livestock
- overfishing in adjacent areas
- waste from nearby settlements and human activities
- pollution, for example oil spills
- changes to local landscapes, for example coastal jetties

▼ **Table 2.6** Mangrove losses (per cent) by cause and region, 2000–16

Region	Approx. amount lost	Loss due to	Loss due to commodities	Loss due to non-productive	Loss due to extreme	Loss due to

	('000 km^2)	settlement		conversion	weather events	erosion
Northern America and the Caribbean	2.0	0	0	8	49	43
South America	0.5	0	17	7	22	54
Africa	0.3	8	7	42	23	20
Asia	2.5	1	68	7	4	20
Oceania	0.5	0	0	8	49	43

- the control of malaria – mangroves are removed as potential breeding grounds for mosquitoes
- increased sea water temperature from power plants located along coastlines, which can damage mangrove ecosystems.

Many of the threats to mangroves come from the demands of the tourism industry and people outside the local area. There are, nevertheless, some benefits of this for local communities, such as food production and erosion control, although overall the negative impacts may outweigh the positive impacts.

Sustainable techniques to promote and manage mangroves

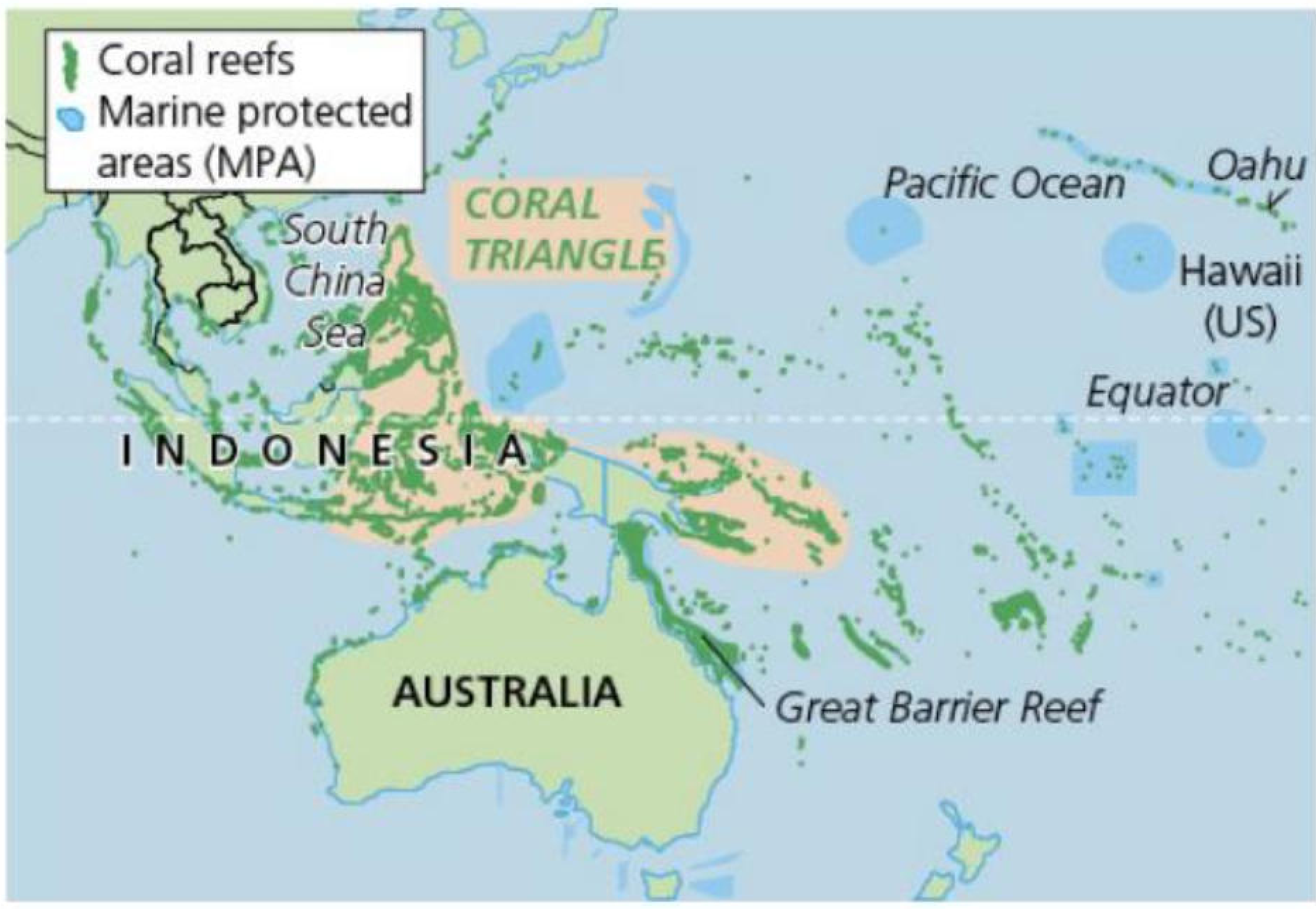
The Zoological Society of London (ZSL), an international conservation charity, has restored over 900 ha of abandoned fishponds in the Philippines back to mangrove forests. Over 50 per cent of mangrove forests in the Philippines had been lost. Over 1 million mangrove trees have been replanted in the Philippines since 2007. Mangroves are now being included in MPAs in Panas, Celen and Bohol in the Philippines. An eco-park featuring mangroves has been established at Katunggan It Ibajay in Panas. Indigenous women have planted over 150,000 mangrove trees on 159 ha of denuded patches of Busuanga Island since 2014. The trees have had an 80 per cent survival rate.

The WWF project 'Mangroves for community and climate aims to protect, restore and strengthen the management of approximately 2.5 million acres of mangroves in Colombia, Fiji, Madagascar and Mexico – 7 per cent of the world's mangroves.

Detailed specific example

The Coral Triangle, Southeast Asia

The richest coral reefs are in the Coral Triangle in Southeast Asia (Figure 2.39). The area includes the waters of Indonesia, Malaysia, Papua New Guinea, the Philippines, Soloman Islands and Timor-Leste. This area of 86,500 km^2 contains two-thirds of the world's coral species, approximately 600 species of reef-building corals, and more than 3000 species of reef fish (twice as many as are found anywhere else), and six of the world's seven sea turtle species.



▲ **Figure 2.39** Coral reefs and marine protected areas (MPAs) in the Coral Triangle

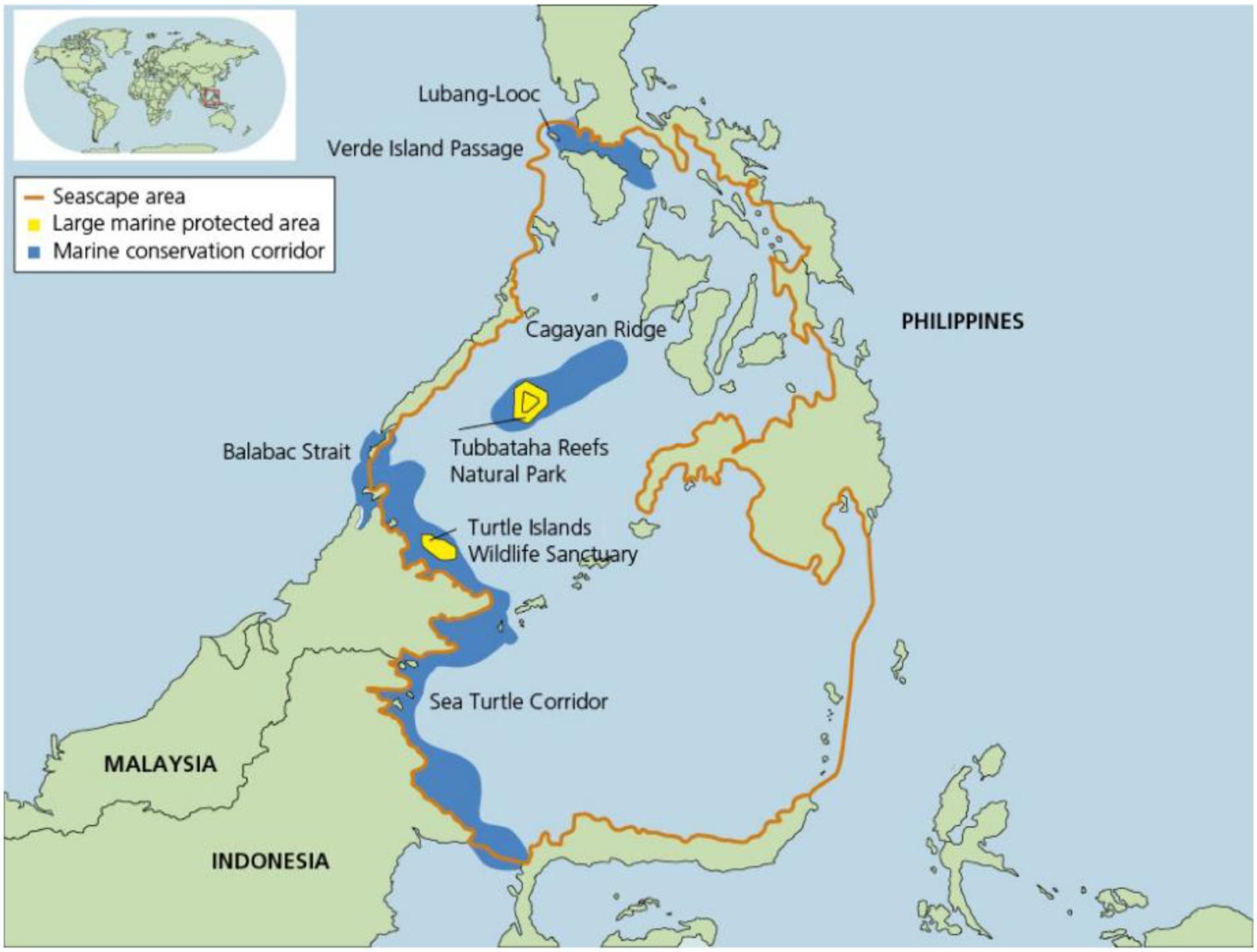
The World Resources Institute estimates that about 60 per cent of reefs face immediate threats. In the past 50 years, about a quarter of all coral cover has died. The reefs that are in the worst shape are those off the most popular beaches, where sun cream gets into the water. In the South China Sea, island building and fishing for giant clams are changing some reefs beyond the possibility of recovery.

The problems vary spatially, hence policies must be locally adapted. The three countries with the largest numbers of people who fish on reefs are all in the Coral Triangle region: Indonesia, Papua New Guinea and the Philippines. In Indonesia and the Philippines, up to 1 million people's livelihoods depend on reefs. Trying to get all these people on board with conservation measures will prove difficult.

One approach is to give ecologically sensitive areas special status, such as that of a **marine protected area (MPA)**. In theory, this means activities that are deemed harmful, such as fishing, drilling and mining, can then be restricted or banned, with penalties for rule-breakers. The Aichi Targets, agreed in 2010 under the UN Convention on Biological Diversity, sought to have at least 17 per cent of inland water and 10 per cent of coastal and marine areas under conservation by 2020. Most countries signed up. As a result of this, about 10 per cent of the Coral Triangle marine habitats are now under effective management.

The most urgent action is regarding fishing vessels. A global register of fishing vessels would help identify wrongdoers. Moreover, simply declaring an area protected does not necessarily protect it. Around 75 per cent of global fisheries come from Asia, and 34 million people there are in engaged in capture fisheries. The industry is worth about US$20 billion annually.

The Coral Triangle initiative has five main goals:



▲ **Figure 2.40** The Sulu-Sulawesi Marine Ecoregion Priority Seascape

- The designation and effective management of priority seascapes such as the Sulu-Sulawesi Marine Ecoregion Priority Seascape (Figure 2.40), which provides a livelihood for around 40 million people, and locally marine managed areas (LMMAs) to protect biodiversity and fish stocks.
- The Ecosystem Approach to Fisheries Management (EAFM), which aims to sustainably use and conserve coastal and marine areas and their natural resources.
- To effectively manage MPAs and other marine resources.
- Adaptation methods to deal with global climate change.
- To improve the status of threatened species such as marine turtles, tuna and blue whales.

Activities

1 Explain the main threats to the Coral Triangle.
2 Describe the methods used to manage the threats to the Coral Triangle.

Fieldwork

Coastal fieldwork

There are many examples of fieldwork that can be carried out in coastal areas, for example sand dune succession, variations in sediment size and shape between the shoreline and the back of a beach, the impact of groynes on beach width, or the impacts of tourism on or development of a coastal area. In all cases, safety is paramount as coastal areas can be dangerous.

Sand dunes are a dynamic environment for several specialised plant species, such as marram grass and sea couch (Figure 2.41). However, sand dunes are increasingly affected by human activities as they are popular places for recreation and tourism, and there are many examples of negative human impacts such as dune erosion, fires and litter. There are also examples of dunes being protected and restored.

Possible hypotheses to test

Possible hypotheses that could be tested in the fieldwork:

- Vegetation density and diversity increase away from the shoreline across the dunes.
- Soil characteristics, such as acidity and organic content, vary with distance from the sea.
- Micro-climate, especially wind speed, varies with distance from the sea.

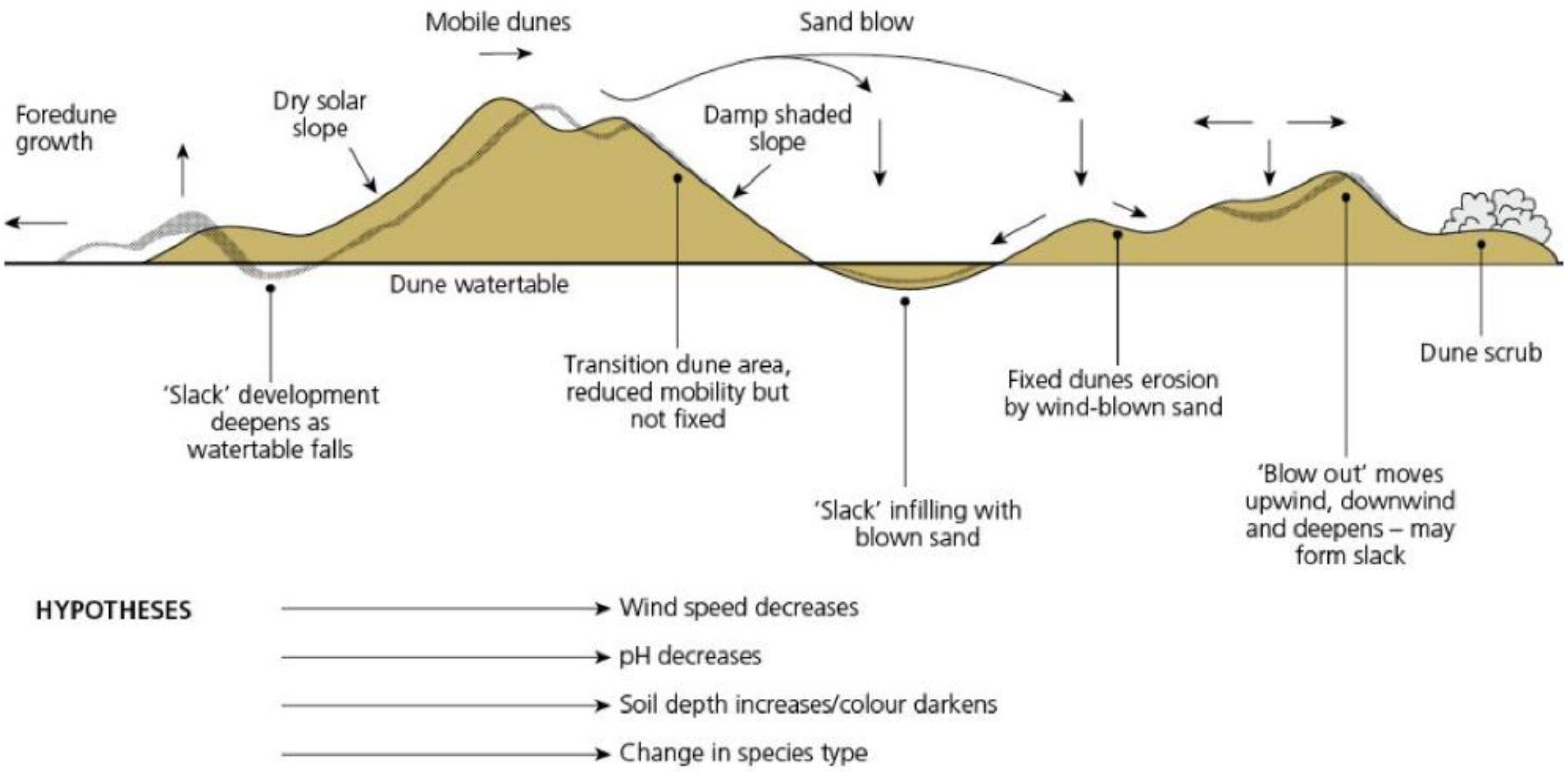
Data collection

Data collection methods include the use of ranging poles, 20 m tapes and clinometers to measure slope length and angle; quadrats to measure vegetation cover; and metre rulers to measure vegetation height. A clinometer can be used to measure slope angle and an anemometer can be used to measure wind speed. A classification sheet of sand dune plants can be used to identify the species present.

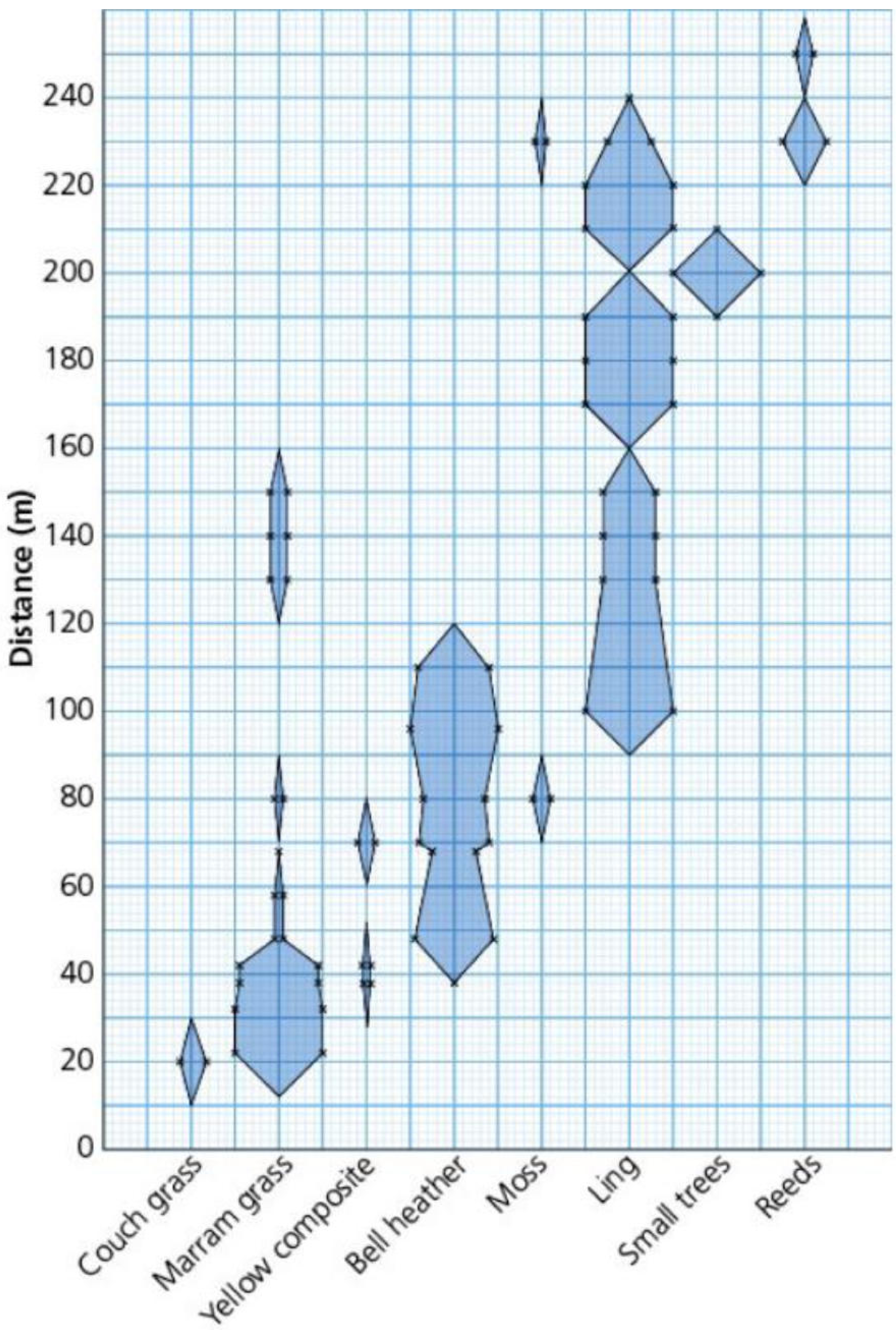
A transect (linear sample) should be taken from the shoreline (or where driftwood first occurs) and samples should be taken at measured intervals.

Presenting your findings

Techniques that could be used to present your findings include a cross-section to show the dune profile, kite diagrams to show vegetation type and density (Figure 2.42), line graphs and scatter graphs to correlate different variables, and statistical tests such as Spearman's rank correlation to test the statistical significance of correlation between variables.



▲ **Figure 2.41** The dynamic sand dune environment



▲ **Figure 2.42** Kite diagram to show percentage cover of sand dune by different species with distance from the sea

Practice questions

1 Figure 2.43 shows a coastal landform.

a Identify the coastal landform. [1]

b Describe the main characteristics of the landform shown. [2]

c Explain how this landform was formed. [3]

▲ **Figure 2.43** A coastal landform

2 **a** Figure 2.44 shows the global distribution of mangrove forests.

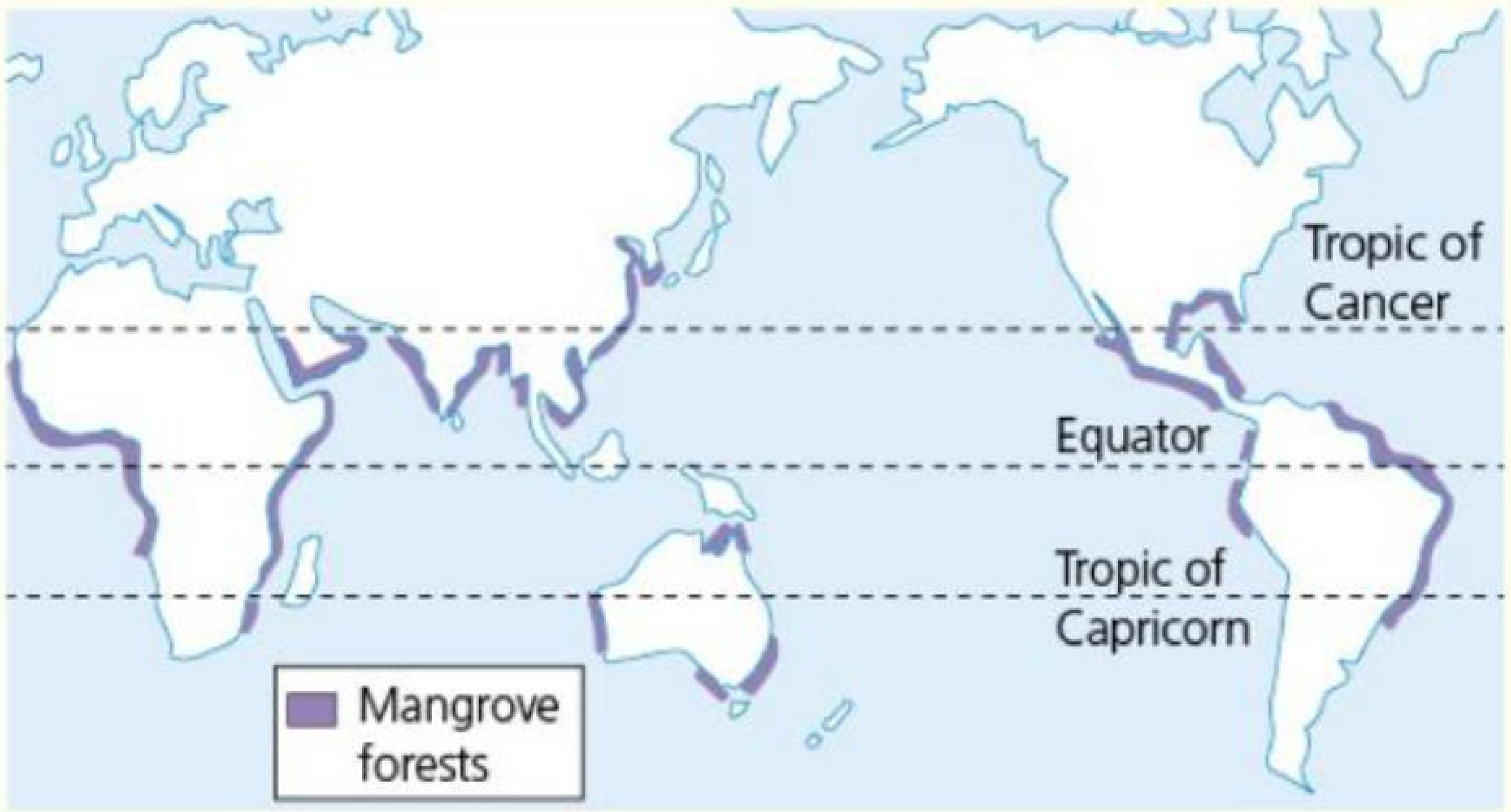
i Describe the distribution of mangrove forests as shown on the map. [3]

ii Explain why mangroves occur in tidal estuaries. [2]

b Figure 2.45 shows part of a mangrove forest at low tide.

i Explain the importance of mangroves in a coastal area that you have studied. [3]

ii Explain one pressure on mangroves. [2]



▲ **Figure 2.44** The global distribution of mangrove forests

▲ **Figure 2.45** A mangrove forest at low tide

3 Assess the opportunities and hazards of living in coastal areas. [7]